



TITLE: Periostat for Periodontal Disease: A Review of the Clinical Effectiveness, Cost Effectiveness and Guidelines

DATE: 10 May 2012

CONTEXT AND POLICY ISSUES

Periodontal disease is a common condition which may be classified as either gingivitis or periodontitis.¹ Patients with gingivitis have gingival inflammation, redness, and bleeding with probing, brushing or flossing. Periodontitis includes gingivitis and loss of supportive tissues including alveolar bone. As a result, there is a loss of attachment of teeth and a probe placed between the gingiva and tooth structures will show increased probing depth. Bone loss around the teeth is detectable on radiographs. Progression of disease may lead to increased tooth mobility and tooth loss.¹

Periodontal disease is closely associated with oral hygiene and is often preventable.¹ Treatment includes good dental hygiene, and scaling and root planing by a dental professional to remove subgingival plaque. Adjunctive therapies include topical or systemic antibiotics. Low dose doxycycline (Periostat 20 mg twice daily) is thought to inhibit collagenase activity and prevent breakdown of connective tissue and alveolar bone in patients with periodontitis.¹

This report will review the evidence on the clinical effectiveness, cost-effectiveness and guidelines for use of doxycycline (Periostat) to inform funding decisions by drug programs.

RESEARCH QUESTIONS

1. What is the clinical effectiveness of Periostat for patients with periodontal disease?
2. What is the cost-effectiveness of Periostat for patients with periodontal disease?
3. What are the evidence-based guidelines for the use of Periostat for patients with periodontal disease?

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KEY MESSAGE

In patients with periodontal disease, doxycycline plus scaling and root planing may improve probing depth relative to scaling and root planing alone, although the clinical importance of the between treatment differences is not clear. No clear benefit with doxycycline was detected in the available studies on other clinical outcomes. No conclusions can be drawn on cost-effectiveness or guidelines for doxycycline use in patients with periodontal disease due to the lack of available evidence.

METHODS

Literature Search Strategy

A limited literature search was conducted on key resources including PubMed, The Cochrane Library (2012, Issue 3), University of York Centre for Reviews and Dissemination (CRD) databases, Canadian and abbreviated list of major international health technology agencies, as well as a focused Internet search. No methodological filters were applied to limit retrieval by study type. Where possible, retrieval was limited to the human population. The search was also limited to English language documents published between January 1, 2007 and April 10, 2012.

Selection Criteria and Methods

One reviewer screened the titles and abstracts of the retrieved publications and evaluated the full-text publications for the final article selection, according to the selection criteria presented in Table 1.

Table 1: Selection Criteria

Population	Patients (any) with periodontal diseases
Intervention	Periostat (doxycycline 20 mg) in combination with scaling and root planing
Comparator	Scaling and root planing alone
Outcomes	Q1. Clinical effectiveness (e.g. improvement in periodontal disease, bone and gum health and recovery, pocket reduction, presence or absence of bleeding) Q2. Cost effectiveness Q3. Guidelines
Study Designs	Q1. HTA, systematic reviews and meta-analyses, RCTs Q2. Economic evaluations Q3. Guidelines

HTA=health technology assessment; RCT=randomized controlled trial

Exclusion Criteria

Studies were excluded if they did not meet the selection criteria, were duplicate publications, or were published prior to 2007.

Critical Appraisal of Individual Studies

RCTs were assessed to determine if allocation to treatment groups was concealed from participants. Randomization, blinding, losses to follow up, and intention to treat (ITT) analysis of data was also assessed.

SUMMARY OF EVIDENCE

Quantity of Research Available

The selection of studies is summarized in Appendix 1. The literature search yielded 108 citations. One additional study was identified by searching the grey literature. After screening of abstracts, 22 potentially relevant studies were selected for full text review. Ten reports on nine RCTs met the inclusion criteria.²⁻¹¹ One systematic review met the inclusion criteria,¹² however upon further examination, the meta-analysis was thought to be inappropriate because it analyzed the data based on probing site without taking into consideration the within person correlation of these data. This would result in confidence intervals that were narrower than the true confidence interval and would therefore be more likely to find statistically significant differences. The systematic review was excluded because of these issues however, two of the three RCTs included in the systematic review have been summarized in this report.^{2,6} The third study did not meet the inclusion criteria because it was published prior to 2007.

No health technology assessments, evidence based guidelines, or economic evaluations met the inclusion criteria.

Summary of Study Characteristics

A total of nine RCTs examined the effects of doxycycline 20 mg twice daily plus scaling and root planing (SRP), compared with placebo plus SRP^{2-6,8,9,11} or SRP alone⁷ in patients with chronic periodontitis. The doxycycline treatment duration was six weeks,^{4,11} three months,^{2,5-8} six months,³ or two years.⁹

One study was a single blind RCT⁷ and all others were double blinded RCTs. The study sample size ranged from 20 to 129 patients, median 34. The two largest studies were conducted in the US,^{7,9} one study was completed in the UK,⁸ and the others were conducted in Turkey^{2,4-6,11} and India.³ The study characteristics are summarized in Appendix 2.

Summary of Critical Appraisal

The critical appraisal of studies is summarized in Appendix 3.

Reporting of allocation concealment was unclear in six RCTs^{2-5,7,11} and randomization methods were not reported in three studies.^{3,4,11} Eight of nine studies evaluated fewer than 25 patients per treatment group,^{2-8,11} and it is possible that despite randomization, prognostic factors were not evenly distributed between groups. Three studies did not complete an intention to treat analysis and excluded patients who were not compliant with therapy or who had missing data.^{2,5,6} The largest study⁹ had more patients in the doxycycline group withdraw from the study than those in the placebo group. In three studies $\geq 20\%$ of patients withdrew or were excluded from the analysis.^{2,5,6} Two studies used the last observation carried forward method for missing data which could potentially bias the findings unless data is missing at random.^{7,8}

Summary of Findings

The key outcomes of the included studies are summarized in Appendix 4 and 5.

Probing depth and clinical attachment level

Probing depth and clinical attachment level (CAL) were reported in all nine RCTs (Appendix 4). Probing depth is the distance from the top of the gingiva (gums) to the point of attachment of the cementum to the tooth root. The CAL is the distance from a fixed point on each tooth, the cement-enamel junction (between the crown and the root), to the base of the probing pocket.⁸ In the RCTs, probing depth and CAL were measured in up to six sites per tooth. For the “whole mouth” analysis, the per-person mean of the probing depth or CAL measurements were used in the analysis. Where possible, the whole mouth data was extracted for this report.

Eight studies reported the probing depth and CAL at baseline and the end of follow-up. In these studies, probing depth and CAL improved over time within both the doxycycline and control groups, and these changes from baseline were statistically significant in many of the trials. The mean probing depth change from baseline was statistically significantly improved in seven RCTs^{2-7,11} within the doxycycline group and in six RCTs^{2-4,6,7,11} for controls (outcome not reported in one RCT⁸). For CAL, the change from baseline was statistically significantly improved in six RCTs^{2,3,5-7,11} and four RCTs^{3,6,7,11} within the doxycycline and control groups respectively (outcome not reported in one RCT⁸).

The difference between groups in probing depth was statistically significantly different favoring doxycycline in five of eight RCTs.^{2-5,11} Two studies calculated the between group difference in the change from baseline.^{3,8} In these studies, the mean improvement in probing depth was 0.3 mm⁸ and 0.5 mm³ higher in the doxycycline than control groups. The difference in CAL between doxycycline and controls was statistically significantly different in one of eight RCTs.³ The between group difference in the change from baseline CAL data was 0.7 mm³ and 0.2 mm⁸ favoring doxycycline.

In the study by Payne et al.,^{9,10} probing depth and CAL were measured at up to 6 sites per tooth and most probing sites (90% to 95%) did not show any change from baseline after two years of treatment. The authors converted the CAL or probing depth change from baseline measurements to three categories: improvement, no change, or disease progression, using the threshold of ≥ 2 mm and ≥ 1.5 mm to define an important change in probing depth and CAL respectively. Patients who received doxycycline were less likely to have progressive worsening of CAL, compared to those who received placebo (odds ratio (OR) 0.81 [95% confidence interval (CI) 0.67 to 0.97]). The odds of progressive worsening in probing depth also favored doxycycline treated patients however the difference was not statistically significant (OR 0.85 [95% CI 0.58 to 1.24]).

Tooth loss

Tooth loss was reported in two RCTs (Appendix 5).^{7,9} Pooled data showed no statistically significant difference between treatment groups on the risk of losing one or more teeth during follow-up (OR 1.13 [95% CI 0.47 to 2.73]).

Alveolar bone loss

One RCT⁹ examined the impact of two year treatment with doxycycline or placebo on alveolar bone density or bone height in postmenopausal women with osteopenia. Most sites (81% to 95%) did not show any change in radiographic outcomes over time.⁹ The odds of progressive loss of alveolar bone density, or alveolar bone height were not statistically significantly different between groups (Appendix 5).⁹

Bleeding on probing

Three RCTs reported bleeding on probing, and in all studies, no statistically significant difference was detected between doxycycline and control groups.^{7,8,10}

Limitations

Most of the RCTs evaluating the clinical effectiveness of doxycycline had methodological limitations that could potentially bias the findings. Reporting of allocation concealment and randomization methods were not clear in many studies. These two elements are important to ensure there is no selection bias or imbalance of prognostic factors when patients are assigned to the treatments. Some studies may have been biased due to exclusion of non-compliant patients, a withdrawal rate >20%, or differential losses to follow-up. Eight of nine studies evaluated small numbers of patient (<25 patients per treatment group). In most studies, the between group differences in probing depth and CAL were calculated using the end of follow-up data, rather than the change from baseline data (preferred analysis).

No relevant evidence-based guidelines for use or economic evaluations were found in the literature search.

CONCLUSIONS AND IMPLICATIONS FOR DECISION OR POLICY MAKING:

Nine randomized controlled trials (RCTs) evaluated the clinical effectiveness of doxycycline 20 mg twice daily plus scaling and root planing, versus scaling and root planing with or without placebo, in patients with periodontal disease. These studies showed that doxycycline may improve the probing depth compared with controls, however, the clinical importance of the between group differences detected (≤ 0.5 mm) is not clear. Most RCTs found no difference between doxycycline and control groups on the clinical attachment level (CAL), although one RCT reported that patients who received doxycycline were less likely to have progressive worsening of CAL compared to placebo. Other outcomes of interest (bleeding on probing, tooth loss, alveolar bone loss) were evaluated in few studies and the impact of doxycycline on these outcomes is not clear.

Most of the RCTs had methodological limitations that may have affected the findings. These limitations included unclear allocation concealment, limited sample size (median 34 patients), lack of intention to treat analysis, and incomplete reporting of randomization methods.

No conclusions can be drawn on the cost effectiveness or guidelines for use of doxycycline in periodontal disease due to the lack of studies.

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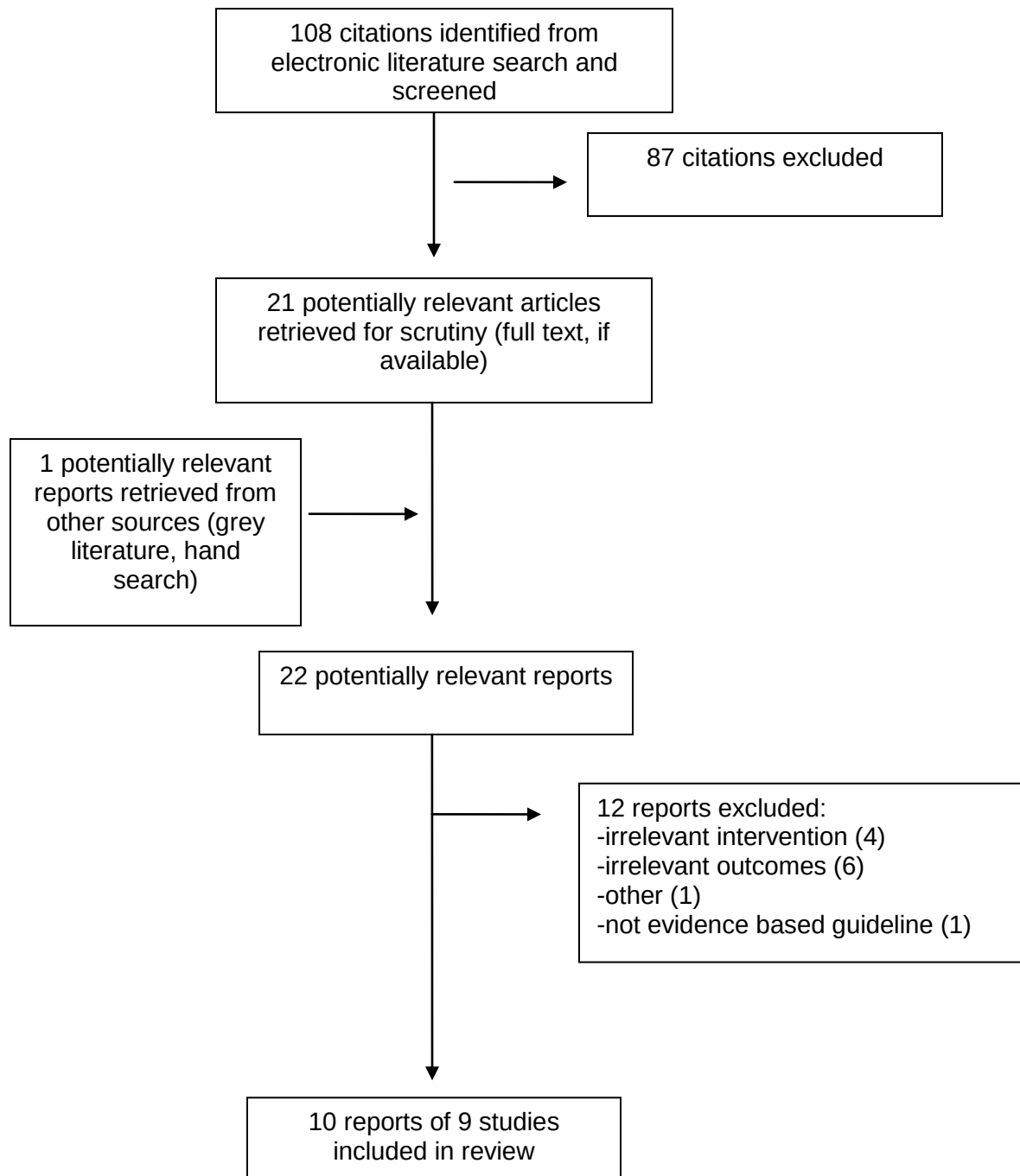
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APPENDIX 1: Selection of Included Studies



APPENDIX 2: Characteristics of Included Studies

Author, Year, Country	Study design, Follow-up	Patient Characteristics, N, Withdrawals	Intervention, Comparator	Key Outcomes
Emingil 2011 ² Turkey	DB RCT 12 months	Moderate to severe chronic periodontitis N=65 Withdrawals: 29%	DOXY 20 mg twice daily + SRP or Placebo + SRP for 3 months	CAL, PD
Deo 2010 ³ India	DB RCT 6 months	Chronic periodontitis and diabetes N=20 Withdrawals: 0%	DOXY 20 mg twice daily + SRP or Placebo + SRP for 6 months	CAL, PD
Tuter 2010 ⁴ Turkey	DB RCT 6 weeks	Chronic periodontitis N=41 Withdrawals: 0%	DOXY 20 mg twice daily + SRP or Placebo + SRP for 6 weeks	CAL, PD
Emingil 2008 ⁵ Turkey	DB RCT 6 months	Chronic periodontitis N=30 Withdrawals: 20%	DOXY 20 mg twice daily + SRP or Placebo + SRP for 3 months	CAL, PD
Gurkan 2007 ⁶ Turkey	DB RCT 12 months	Severe chronic periodontitis N=35 Withdrawals: 26%	DOXY 20 mg twice daily + SRP or Placebo + SRP for 3 months	CAL, PD
Haffajee 2007 ⁷ US	SB RCT 12 months	Chronic periodontitis (>20 years of age) N=98 Withdrawals: 6%	DOXY 20 mg twice daily for 12 weeks + SRP or Azithromycin 500 mg daily x 3 days + SRP or Metronidazole 250 mg three times daily X 14 days + SRP or SRP alone	CAL, PD, BOP, tooth loss
Needleman 2007 ⁸ UK	DB RCT 6 months	Chronic periodontitis and current smoker (age 30 to 70 years) N=34 Withdrawals: 12%	DOXY 20 mg twice daily + SRP or Placebo + SRP for 3 months	CAL, PD, BOP
Payne 2007 ^{9,10} US	DB RCT 2 years	Post-menopausal osteopenic women (age 45 to 70 years)	DOXY 20 mg twice daily or Placebo for 2 years	Alveolar bone loss, tooth loss,

Author, Year, Country	Study design, Follow-up	Patient Characteristics, N, Withdrawals	Intervention, Comparator	Key Outcomes
		with moderate to severe periodontitis N=129 Withdrawals: 9%	Both groups underwent periodontal maintenance therapy every 3 to 4 months; daily calcium and vitamin D supplements	BOP, CAL, PD
Tuter 2007 ¹¹ Turkey	DB RCT 6 weeks	Chronic periodontics and coronary artery disease N=36 Withdrawals: 0%	DOXY 20 mg twice daily + SRP or Placebo + SRP for 6 weeks	CAL, PD

BOP=bleeding on probing; DB=double blind; CAL=clinical attachment level; DOXY=doxycycline; PD=probing depth; RCT=randomized controlled trial; SRP=scaling and root planing

APPENDIX 3: Validity Assessment of Included Studies

Author, Year	Strengths	Weaknesses
Emingil 2011 ²	- Random allocation, double blind	- Allocation concealment not clear - Not ITT: Analyzed compliant patients (71%) - Limited sample size (N=46)
Deo 2010 ³	- Random allocation, double blind	- Allocation concealment and randomization methods were not reported - Limited sample size (N=20) - Incomplete reporting of patient characteristics and study outcomes
Tuter 2010 ⁴	- Random allocation, double blind - All patients completed study	- Allocation concealment, randomization and blinding methods not reported - Limited sample size (N=41)
Emingil 2008 ⁵	- Random allocation, double blind	- Allocation concealment unclear. - Limited sample size (N=24) with some imbalances between groups at baseline - Not ITT: Analyzed patients with complete data (80%)
Gurkan 2007 ⁶	- Random allocation, double blind	- Sample size limited (N=26) - Not ITT: Analyzed compliant patients (74%)
Haffajee 2007 ⁷	- Random allocation - ITT analysis	- Allocation concealment unclear. - Single blinded (not clear who was masked) - LOCF for missing data - Sample size limited (20 to 25 patients per group)
Needleman 2007 ⁸	- Used adequate randomization, allocation concealment and blinding methods. - ITT analysis	- Limited sample size (N=34). - LOCF for missing data
Tuter 2007 ¹¹	- Random allocation, double blind	- Allocation concealment and randomization methods not reported - Sample size limited (N=36)
Payne 2007 ^{9,10}	- Used adequate randomization, allocation concealment and blinding methods. - ITT analysis - The analysis of data accounted for the within person correlation of the outcome measurements.	- Fewer patients in the DOXY group completed the study than in the placebo group (78% versus 97%).

DOXY=doxycycline; ITT=intention to treat; LOCF=last observation carried forward

APPENDIX 4: Results for Pocket Depth and Clinical Attachment Level

Author, year, treatment/follow-up duration	Group	N	Mean PD (mm)		PD between group difference P value	Mean CAL (mm)		CAL between group difference P value
			baseline	end of follow-up		baseline	end of follow-up	
Emingil 2011 ² 3/12 months	DOXY	23	4.0	2.4	P<0.05	4.7	4.2	NS
	placebo	23	4.2	2.9		5.2	4.8	
Deo 2010 ³ 6/6 months	DOXY	20	5.7	2.6	0.5 mm*	6.1	3.8	0.7 mm*
	placebo		5.7	3.2	p<0.001	6.0	4.5	P<0.001
Tuter 2010 ^{4**} 6/6 weeks	DOXY	20	4.3	3.1	P=0.002	4.1	3.8	NS
	placebo	21	4.2	3.4		4.1	3.5	
Emingil 2008 ⁵ 3/6 months	DOXY	12	3.7	2.3	P<0.05	4.0	3.7	NS
	placebo	12	3.9	2.6		4.7	4.6	
Gurkan 2008 ⁶ 3/12 months	DOXY	13	4.6	3.0	NS†	5.7	4.8	NS
	placebo	13	4.3	3.1		5.7	5.0	
Haffajee 2007 ⁷ 3/12 months	DOXY	20	3.3	3.0‡	NS	3.5	3.2‡	NS
	control	23	2.9	2.6‡		3.0	2.9‡	
Needleman 2007 ⁸ 3/6 months	DOXY	16	6.0‡	4.7‡	0.3 mm* (95% CI -0.1 to 0.7) NS	6.3‡	5.7‡	0.2 mm* (95% CI -0.2 to 0.6) NS
	placebo	18	5.8‡	4.8‡		5.7‡	5.2‡	
Tuter 2007 ^{11**} 6/6 weeks	DOXY	18	4.4	3.5	P=0.03	4.1	4.0	NS
	placebo	18	4.6	3.8		4.3	4.0	

CAL=clinical attachment level; CI=confidence interval; DOXY=doxycycline; NR=not reported; NS=not statistically significant; PD=probing depth

*Between group difference in the change from baseline. Positive numbers indicate that the DOXY group improved more than the placebo group.

**Data reported as median PD or CAL in mm

†No statistically significant difference between groups was detected in the PD or CAL change from baseline results for subgroups with baseline PD or CAL measurements of 4 to 6 mm, or ≥7mm in depth.

‡Estimated from figures.

Results for Pocket Depth and Clinical Attachment Level from Payne et al.

Author, year, Treatment/follow-up duration	N	Outcome	OR (95% CI) DOXY versus placebo
Payne 2007 ¹⁰ 2/2 years	128	Progressive change in PD*	0.85 (0.58 to 1.24) P=NS
		Progressive change in CAL*	0.81 (0.67 to 0.97) P=0.03

CAL=clinical attachment level; CI=confidence interval; DOXY=doxycycline; NS=not statistically significant; PD=probing depth

*PD and CAL measurements were converted to a categorical outcome: progression, no change, or improvement. Disease progression was defined as an increase in CAL by ≥ 1.5 mm or increase in PD by ≥ 2 mm from baseline.

APPENDIX 5: Results for Other Outcomes

Tooth Loss

Author, year	Outcome	DOXY n/N (%)	Control n/N (%)	OR (95% CI) DOXY versus control
Haffajee 2007 ⁷	Loss of 1 or more teeth over 1 year	2/20 (10%)	0/23 (0%)	NE
Payne 2007 ⁹	Loss of 1 or more teeth over 2 years	9/51 (18%)	12/62 (19%)	0.89 (0.34 to 2.32) P=NS†

CI=confidence interval; DOXY=doxycycline; NE=not estimable; NS=not statistically significant; OR=odds ratio

†calculated by CADTH. Pooled data from both studies using fixed effect meta-analysis: OR 1.13 [95% CI 0.47 to 2.73] P=0.78.

Alveolar Bone Loss

Author, year	Outcome	OR (95% CI) DOXY versus placebo
Payne 2007 ⁹	Progressive loss of alveolar bone density over 2 years	RA: 1.04 (0.80 to 1.34) P=NS CADIA: 0.84 (0.65 to 1.08) P=NS
	Progressive alveolar bone height loss over 2 years	0.82 (0.62 to 1.08) P=NS

CADIA=computer assisted densitometric image analysis; CI=confidence interval; DOXY=doxycycline; NS=not statistically significant; OR=odds ratio; RA=radiographic absorptiometry

Bleeding on Probing

Author, year	Bleeding on probing
Haffajee 2007 ⁷	Change from baseline similar in the DOXY and SRP groups
Needleman 2007 ⁸	Change from baseline, DOXY versus placebo 2.0% (95% CI -8.1% to 12.2%) P=NS*
Payne 2007 ¹⁰	OR 0.82 (95% CI 0.61 to 1.10) P=NS

CI=confidence interval; DOXY=doxycycline; OR=odds ratio; SRP=scaling and root planing

*Change in subject-level % of sites bleeding on probing for DOXY group versus placebo at 6 months. Positive numbers indicate improvement in bleeding on probing.